Név: .... **NEPTUN:** .... Score: ☐ Grade: ☐

Fundamentals of thery of computation I. – sample

0-23: grade 1, 24-31: grade 2, 32-39: grade 3, 40-47: grade 4, 48-60: grade 5

1. True or false? Write the first letter of your answer (T or F) into the box. (20x1 pont)

- ✗ ab is a suffix of $abbaba$.
- ✓ $L\{\varepsilon\} = L$ holds for every language $L \subseteq \{a, b\}^*$.
- ✗ Every language that can be generated by a linear grammar can be described by a regular expression.
- ✗ The regular expression cc^* describes the language $\{c^{2n} \,|\, n \geq 0\}$.
- ✓ Every language that can be described by a regular expression can be generated by a type 3 normal form grammar.
- ✓ Let $A = (Q, \Sigma, \delta, Q_0, F)$ be a nondeterministic finite automaton. Then there is a regular expression R such that $L(R) = L(A)$ holds.
- ✗ $G = (\{S, X\}, \{c, d\}, \{S \to cX, S \to \varepsilon, X \to ddX, X \to cd\}, S)$ is a grammar of type 3 normal form.
- ✓ Let $G = (N, \Sigma, P, S)$ be a length-nondecreasing grammar. If $u \to v \in P$ and $u \neq S$ hold, then $|v| \geq 1$.
- ✗ If all rules of a grammar $G = (N, \Sigma, P, S)$ is of the form $A \to BC$, $AB \to AC$, $BA \to CA$, $A \to a$ or $A \to \varepsilon$ $(A, B, C \in N, a \in \Sigma)$ then G is a grammar of Kuroda normal form.
- ✓ $\{u \in \{a, b\}^* \,|$ the number of b's is odd in $u\}$ is a context-free language.
- ✓ The membership problem of type 2 grammars is decidable in polynomial time.
- ✗ If $B \in N$ is a reachable nonterminal of a context-free grammar $G = (N, \Sigma, P, S)$ then there exists a nonterminal $A \in N$ such that $A \to B \in P$ holds.
- ✗ Let $A = (Q, \Sigma, \delta, q_0, F)$ be a deterministic finite automaton and let $q \in Q$ be a state reachable from q_0. Then there exists a word $w \in \Sigma^*$ with the property $q_0 w \Rightarrow q$.
- ✓ The class of regular languages is closed under the operation intersection.
- ✓ The class of context sensitive languages is closed under the operation union.
- ✓ Let $A = (Z, Q, \Sigma, \delta, z_0, q_0, F)$ be a pushdown automaton. Then $L(A)$ is of type 2.
- ✗ Let $A = (Z, Q, \Sigma, \delta, z_0, q_0, F)$ be a pushdown automaton and let $(x, r) \in \delta(z, q, b)$ be a transition of A $(x, z \in Z, b \in \Sigma, q, r \in Q)$. Applying this transition the stack will contain one more letter then in the previous configuration.
- ✓ Let $A = (Z, Q, \Sigma, \delta, z_0, q_0, F)$ be a pushdown automaton. Then there is a grammar G in Chomsky normal form with the property $L(G) = L(A)$.
- ✓ Let $G = (N, \Sigma, P, S)$ be a grammar in Chomsky normal form and consider a derivation tree t of a word from $L(G)$. Then all nodes of t have 0, 1 or 2 children.
- ✗ It is possible to prove that a given language L belongs to the class of context-free languages by applying Bar-Hillel Lemma for L.

NEPTUN:.........................

2. Answer the following questions. No explanation required. (10x2 pont)

 (a) Give a formula describing the regular expression a^*b^*.

 $\{a^n b^k \mid n, k \geq 0\}$

 (b) What language operations are called regular operations?

 union, concatenation, iterative closure

 (c) Let $G = (N, \Sigma, P, S)$ be a grammar and let $u, v \in (N \cup \Sigma)^*$. Define what does it mean, that v is derivable from u. ($u \Rightarrow_G^* v$)

 either $u = v$ or there exist an integer $n \geq 1$ and words $w_0, \ldots w_n \in (N \cup \Sigma)^*$, such that $w_{i-1} \Rightarrow_G w_i$ $(1 \leq i \leq n)$, $w_0 = u$ and $w_n = v$

 (d) Let $A = (Q, \Sigma, \delta, Q_0, F)$ be a nondeterministic finite automaton. Define $L(A)$.

 $L(A) = \{u \in \Sigma^* \mid q_0 u \Rightarrow_A^* p$ holds for some $q_0 \in Q_0$ and $p \in F\}$

 (e) Consider a nondeterministic finite automaton $A = (\{p, q, r, s\}, \Sigma, \delta, \{p, r\}, \{p, q\})$ (i.e., $F = \{p, q\}$). List all the states in $Q' \setminus F'$, where $A' = (Q', \Sigma, \delta', q_0', F')$ is the deterministic finite automaton with $Q' = P(Q)$ we have learnt to be equivalent with A.

 $\emptyset, \{r\}, \{s\}, \{r, s\}$

 (f) List all the suffix languages with respect to the the words over the alphabet $\{a, b\}$ for the language $L = \{a, bb\}$.

 $\emptyset, \{\varepsilon\}, \{b\}, \{a, bb\}$

 (g) Assume, that the following sets are already known for an input of the CYK algorithm.

 $H_{22} = \{A, C\}, \quad H_{33} = \{B\}, \quad H_{44} = \{B\}$ and $H_{23} = \{B\}$, $H_{34} = \{C\}$.

 Furthermore assume, that the rules of the grammar having only A, B and C on the right hand side are the following.

 $C \to CA \mid BC, \quad D \to CC \mid BB, \quad E \to BB.$

 Then $H_{24} = \{D, E\}$

 (h) Give a pushdown automaton rule with the following effect. The automaton processes an input letter b, removes a letter c from the pusdown store and changes the state of the automaton from q_3 to q_1.

 Solution 1: $cq_3b \to q_1$ Solution 2: $(\varepsilon, q_1) \in \delta(c, q_1, b)$

 (i) Place $\mathcal{L}_{\text{detPDA}}$, the class of languages being recognizable by a deteministic pushdown automaton in Chomsky's hierarchy of language classes.

 $\mathcal{L}_3 \subset \mathcal{L}_{\text{detPDA}} \subset \mathcal{L}_2$ (strict inclusions!)

 (j) Define the transition function δ of the pushdown automaton $A = (Z, Q, \Sigma, \delta, z_0, q_0, F)$.

 $\delta : Z \times Q \times (\Sigma \cup \{\varepsilon\}) \to \mathcal{P}_{\text{fin}}(Z^* \times Q)$

NEPTUN:..........................

3. (3+4+7+6 points)

(a) Let $G = (N, \Sigma, P, S)$ be a grammar of type 2. Give the construction of a grammar of type 2 generating $L(G)^*$ according to the construction we have learnt.

$G = (N, \Sigma, P \cup \{S \to SS \,|\, \varepsilon\}, S)$

(b) Define what is meant by a grammar in Chomsky normal form and state the normal form theorem with respect to it.

A grammar $G = (N, \Sigma, P, S)$ is in *Chomsky normal form*, if all rules of P has one of the following formats.

- $S \to \varepsilon$
- $A \to BC$, $A, B, C \in N$; furthermore $B, C \neq S$ should hold in the case $S \to \varepsilon \in P$.
- $A \to a$, $A \in N, a \in \Sigma$.

Theorem For every context-free grammar $G = (N, \Sigma, P, S)$ there exist a grammar $G = (N', \Sigma, P', S')$ in Chomsky normal form with the property $L(G') = L(G)$.

(c) Present the length reduction algorithm we have learnt for regular grammars. Give the format of the rules before and after the length reduction.

Input: all rules are of the following form.

a) $A \to uB$ $(A, B \in N, u \in \Sigma^*)$ or
b) $A \to u$ $(A, B \in N, u \in \Sigma^*)$.

for rules of type a): If $A \to uB \in P$ $(A, B \in N, u \in \Sigma^*)$, $|u| > 1$ and $u = a_1 \cdots a_n, n \geq 2$, then replace the rule $A \to a_1 \cdots a_n B$ by the set of rules $\{A \to a_1 Z_1, Z_1 \to a_2 Z_2, \ldots Z_{n-1} \to a_n B\}$, where $Z_1, \ldots, Z_{n-1}$ are new nonterminals, they can not be reused for the reduction of other long rules.

for rules of type b): Similarly, if $A \to u \in P$ $(A \in N, u \in \Sigma^*)$, $|u| > 0$ and $u = a_1 \cdots a_m, m \geq 1$, then replace the rule $A \to a_1 \cdots a_m$ by the set of rules $\{A \to a_1 Y_1, Y_1 \to a_2 Y_2, \ldots Y_{m-1} \to a_m E, E \to \varepsilon\}$ where $Y_1, \ldots, Y_{m-1}$ are new nonterminals, they can not be reused for the reduction of other long rules. E should be a new nonterminal as well, but it can be used for all the length reductions of type b) rules.

Output: all rules of the grammar after length reducion are of one the following forms.
$X \to aY, X \to Y, X \to \varepsilon \quad (X, Y \in N, a \in \Sigma)$.

(d) Define what is meant by a suffix language with respect to a state $q \in Q$ of a deterministic finite automaton $A = (Q, \Sigma, \delta, q_0, F)$.

Give a system of linear language equations of size $|Q|$, where the unknowns are these suffix languages.

$L(A, q) = \{v \in \Sigma^* \,|\, \delta(q, v) \in F\}$

(Note: $L(A, q) = \{v \in \Sigma^* \,|\, qv \Rightarrow_A^* p, p \in F\}$ is another good solution.)

The following linear equations on the suffix languages hold.

$$L(A, q) = \bigcup_{t \in \Sigma} \{t\}\, L(A, \delta(q, t)) \cup \begin{cases} \emptyset & \text{if } q \notin F \\ \{\varepsilon\} & \text{if } q \in F \end{cases}$$